Pattern-Like Astrology App

Product, UX, Data, and Platform Design Specification

Version	0.2
Date	July 30, 2026
Status	Product and engineering baseline
Calculation authority	Swiss Ephemeris
Architecture profile	Cloudflare-first; WordPress.com editorial; Fly.io portable compute

Governing boundary

Celestial calculations establish eligibility. Personal context may rank, frame, prompt, or schedule a valid interpretation, but it may never alter chart facts or be presented as something astrology independently discovered.

Document scope

Product definition, experience architecture, astrological contract, complete automatable-source governance, Cloudflare/WordPress.com/Fly.io deployment design, data model, workflows, API, security, reliability, and implementation gates.

Revision summary

This document supersedes the v0.1 Automatable Data Source Specification while preserving its 139-entry registry as a normative companion artifact.

Area	v0.2 addition
Product and UX	Primary navigation, screen requirements, evidence interactions, notifications, and visual direction.
Platform	Concrete Cloudflare-first topology with WordPress.com editorial workflows and Fly.io portability.
Content operations	REST-enabled WordPress content model, signed releases, validation, activation, and rollback.
Data and security	D1/R2 assignment, envelope encryption, crypto-shredding, service boundaries, and AI privacy.
Operations	Durable workflows, queue idempotency, SLOs, observability, migration triggers, and acceptance criteria.
Registry	Each source now has an implementation owner, primary runtime, storage targets, and delivery path.

Contents

1. Executive architecture decision

Recommended launch profile

Run the product on Cloudflare. Use WordPress.com as the editorial control plane. Keep Fly.io deployment-ready, but outside the launch critical path unless measured latency, batch, or database requirements justify it.

This split minimizes operational surfaces while preserving portability for the native Swiss Ephemeris engine and a future relational database.

Platform	Launch role	Required?	Boundary
Cloudflare	Primary runtime, orchestration, data, objects, cache, calculation, and operational telemetry.	Yes	Private content stays out of analytics; model logs off for private synthesis.
WordPress.com	Editorial authoring, review, release packaging, public help and policy content.	Yes	No user birth, journal, relationship, health, or private reading data.
Fly.io	Portable native compute, warm pool, backfills, and optional Managed Postgres.	Deployment-ready	Not introduced into request flow without a documented trigger.
Mobile clients	User experience, device permissions, local aggregation, secure local cache.	Yes	Transmit only permitted/minimized source representations.

Launch decisions

- Use Cloudflare Containers for Swiss Ephemeris at launch. Bake the library, data files, configuration, and golden-fixture runner into one signed OCI image.
- Keep the same image runnable on Fly Machines. Portability is tested in CI even while Cloudflare remains primary.
- Use D1 for launch operational records with application-layer envelope encryption.
- Use R2 for signed editorial bundles, chart renders, encrypted exports, and immutable calculation artifacts.
- Use deterministic interpretation assembly by default; constrained model editing is optional and never required for a valid reading.
- Use WordPress.com for reviewed interpretation content and release governance, not for application user data.

2. Product definition

A private psychological-timing app that identifies stable natal patterns, calculates active developmental cycles with Swiss Ephemeris, and presents one grounded daily interpretation as part of a visible longer-term timeline.

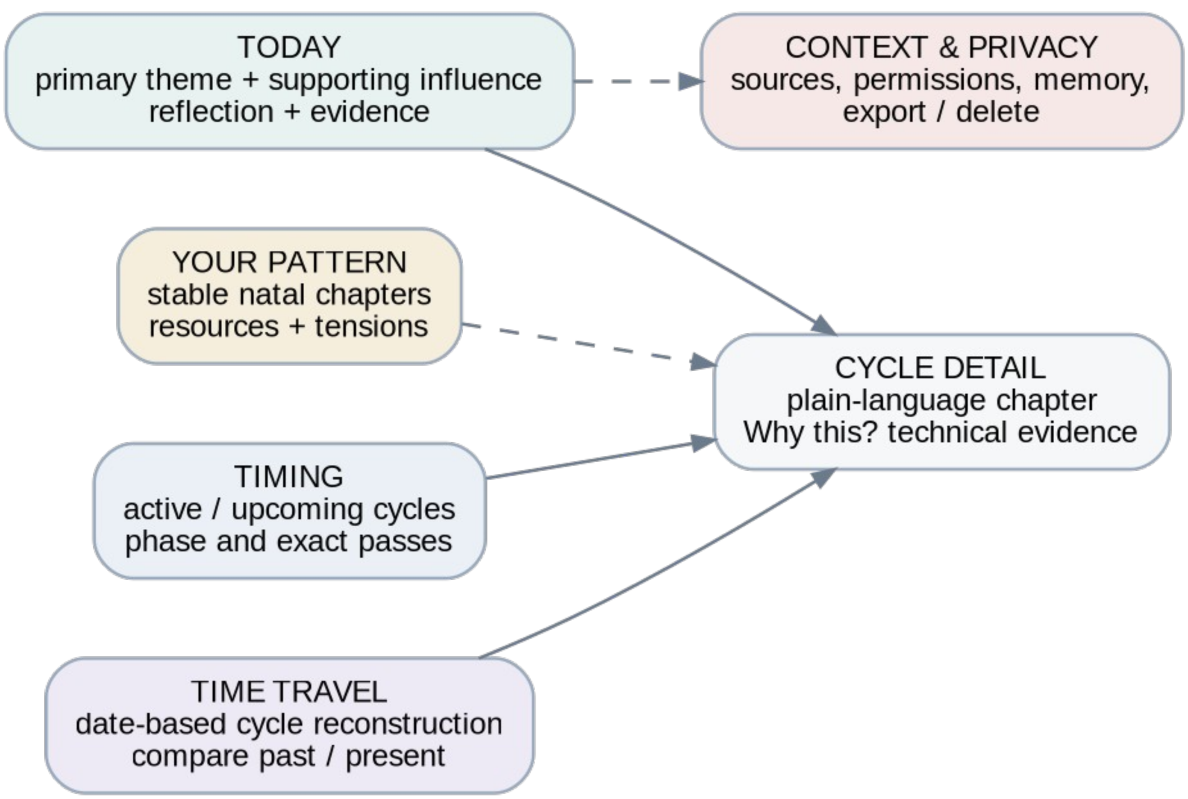
Product promise

- Pattern-like depth: integrated psychological chapters rather than isolated placements.
- Daily usefulness: one coherent theme, one supporting influence at most, and one reflection.
- Inspectable reasoning: every reading exposes its cycle, phase, timing, uncertainty, and content lineage.
- Private by design: context remains optional, minimized, reversible, and visibly distinct from astrology.

Non-goals

- No deterministic claims about external events, health, pregnancy, death, crime, money, or another person's intentions.
- No engagement-maximizing fear notifications or endless Moon-transit noise.
- No hidden surveillance integrations, wholesale contact ingestion, or private-message mining.
- No user-data dependency on WordPress.com and no LLM-calculated chart facts.

Primary navigation



Experience map: plain-language surfaces share a common technical evidence layer.

3. Experience design

Surface	Primary job	Required content	Key interaction	Uncertainty/failure behavior
Today	Give the user one useful daily chapter.	Primary theme, optional support, reflection, phase, time range.	Save, reflect, expand Why this?	Never blank; deterministic fallback; label missing-time limitations.
Cycle detail	Explain the full developmental arc.	Plain-language chapter, phase variants, exact passes, resources, tensions.	Move through passes; open related readings.	Show last calculation, contract, and uncertainty.
Your Pattern	Present stable natal patterns as an integrated profile.	20-30 reviewed families at launch, strengths, tensions, counter-expressions.	Open evidence and related cycles.	Do not fill missing evidence with generic sign copy.
Timing	Show active and approaching cycles.	Emerging, building, peak, reconsidering, integrating, upcoming.	Filter by domain/duration.	Expose stale calculation state; allow refresh.
Time Travel	Reconstruct a selected date.	Cycle state, saved context, related readings, comparison to now.	Choose date; compare periods.	Differentiate historical facts, saved user context, and retrospective synthesis.
Context & Privacy	Make personalization controllable.	Sources, scope, freshness, memory, AI policy, export, delete.	Pause, revoke, reset, delete.	Revocation is immediate for future jobs; show completion state.

Visual system

Token	Specification	Purpose
Character	Quiet, intimate, precise, non-predictive.	Avoid cosmic cliches and gamified urgency.
Primary	#18334D midnight navy	Structure, headings, evidence.
Accent	#2E7477 muted teal	Active controls and trustworthy status.
Warm accent	#B98A3D muted gold	Cycle milestones and exact-pass markers.
Surface	#F5F7F8 / warm white	Low-contrast, readable content planes.
Interpretation type	Readable serif; 18-24 px mobile	Long-form reflective copy.
Control type	Humanist sans; 15-17 px mobile	Navigation and actions.
Evidence type	Monospace; 12-14 px mobile	Fact IDs, dates, orbs, versions.
Motion	Short fades and progress movement only	Express cycle phase without spectacle.

4. Astrological and interpretation contract

Zodiac	Tropical
Coordinates	Geocentric, ecliptic-of-date, apparent positions, speed enabled
Houses	Placidus primary; Porphyry fallback recorded when required
Node	True lunar node
Launch bodies	Sun through Pluto, true node, Ascendant, Midheaven
Launch aspects	Conjunction, sextile, square, trine, opposition
Calculation authority	Pinned Swiss Ephemeris release, data files, flags, wrapper commit, and container digest

Birth-time modes

Mode	Allowed	Disabled/qualified
Exact	Full chart, houses, angles, angle transits, high-confidence Moon.	Only technique-specific uncertainty.
Approximate	Features stable across the declared uncertainty window.	Any house/angle/Moon feature that changes across the interval.
Unknown	Planetary signs and stable aspects; slow transits.	Houses, angles, angle transits, and time-sensitive Moon claims.

Cycle lifecycle

Calculated state	Product phase	Narrative obligation
Entered orb	Emerging	Introduce the question without implying an event.
Applying	Building	Describe increasing relevance or pressure.
Near exact	Peak	Name the clearest expression; avoid alarm.
Retrograde return	Reconsidering	Frame recurrence as revision, not destiny.
Final separating pass	Integrating	Emphasize learning, choice, and what remains.

Calculation separation

Swiss Ephemeris calculates positions and timing. Application rules derive aspects and cycles. Editorial content interprets eligible structures. An LLM, if enabled, may only edit approved material.

5. Automatable data-source model

The v0.2 registry contains 139 entries. It preserves every feasible source class and every deliberate exclusion from v0.1, then maps each entry to an implementation owner, runtime, storage target, and delivery path.

Category	Entries	Default owner	Primary runtime	Primary storage
A. Calculation authority and astrological sources	20	profile-service	cloudflare-api-worker	d1: encrypted operational record; r2: immutable snapshot or rendered artifact
B. First-party, user-owned sources	24	first-party-context-service	cloudflare-api-worker + workflow	d1: encrypted structured record
C. Device and operating-system sources	17	device-permission-adapter	mobile-client; cloudflare-api-worker receives minimized aggregate	on-device first; d1: encrypted aggregate only when allowed
D. Linked cloud-service sources	20	linked-service-adapter	cloudflare-connector-worker + workflow	d1: encrypted normalized aggregate; secret store: provider credentials/tokens where applicable
E. Public and ambient sources	12	ambient-data-adapter	cloudflare-connector-worker	kv/cache: short-lived lookup; d1: evidence snapshot only when used
F. Derived and inferred features	22	reading-intelligence-service	cloudflare-workflow + deterministic rules package	d1: derived feature and evidence linkage
G. Deliberately excluded sources and uses	24	policy-gate	none	None

Evidence lanes and permission tiers

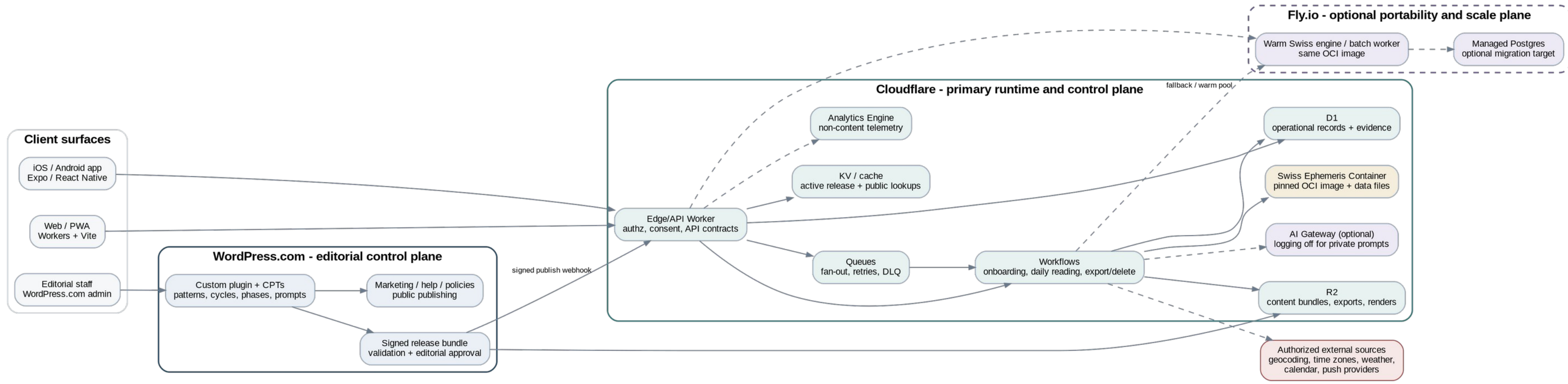
Lane/tier	Default	May influence	Must never do
Celestial facts / Tier 0	Required	Chart facts, cycle timing, eligibility.	Diagnose, guarantee an event, or claim causation.
First-party reflection / Tier 1	Explicit controls	Domain selection, prompts, continuity, ranking.	Become hidden inference or mutate chart facts.
Situational context / Tier 2	Optional opt-in	Timing, practical framing, environment.	Pretend to be an astrological prediction.
Wellness/behavior / Tier 3	Off; separate consent	Only low-resolution tone/timing adjustments.	Create health claims or enter raw model prompts.
Prohibited / Tier 4	Never	Nothing.	Ingest, train, target, diagnose, or make eligibility decisions.

Launch source classes

Launch source class	Entries
A. Calculation authority and astrological sources	12
B. First-party, user-owned sources	19
C. Device and operating-system sources	6
E. Public and ambient sources	4
F. Derived and inferred features	13

Launch coverage includes calculation authority, explicit user context, minimized device delivery context, ambient timing, and deterministic derived features. Exact source IDs, permissions, storage routes, and allowed uses are normative in pattern_like_astrology_app_data_source_registry_v0.2.yaml.

6. Platform architecture



Recommended deployment: Cloudflare primary, WordPress.com editorial, Fly.io optional portability and scale plane.

Service responsibility matrix

Component	Platform/service	Responsibility	State/data boundary	Launch
Mobile app	Expo / React Native	Core experience, permissions, on-device minimization, secure cache.	No provider secret; upload only allowed aggregates.	Required
Account web	Cloudflare Workers + Vite	Settings, evidence, export/delete, support.	Same API and authorization policy as mobile.	Required
Edge API	Cloudflare Worker	Contracts, authz, consent, idempotency, rate limiting.	No raw private payloads in logs.	Required
Orchestration	Workflows + Queues	Durable chart/reading/release flows, fan-out, retries, DLQ.	Serializable state only; consumers are idempotent.	Required
Calculation	Cloudflare Container	Pinned Swiss Ephemeris service and exact-search engine.	Immutable image digest and data-file hashes.	Required
Data plane	D1 + R2 + KV/cache	Encrypted records, evidence, releases, exports, immutable artifacts.	Envelope encryption; cache contains no raw private content.	Required
Editorial control	WordPress.com	Author, review, approve, sign, release, publish help/policies.	No application user data.	Required
Optional model route	AI Gateway	Provider routing, budgets, fallbacks, rollback.	Private logging disabled; no stable user identifier.	Optional
Portability and scale	Fly Machines + Managed Postgres	Warm compute, backfills, future relational migration.	Same OCI image; private service boundary; not launch-critical.	Standby

7. WordPress.com editorial control plane

WordPress.com is the human content system. It provides familiar roles, revisions, editorial review, media, and REST-extensible custom content while remaining isolated from user application data.

Custom content types

Content type	Purpose	Required fields	REST/release behavior
astrology_pattern	Stable natal pattern family.	Title, evidence requirements, resources, tensions, counter-expression, prompts.	REST enabled; immutable version in release.
astrology_cycle	Transit/progression/return interpretation.	Technique, bodies/targets, aspect, orb policy, themes, prohibited claims.	REST enabled; links to phase variants.
astrology_phase	Emerging/building/peak/reconsidering/integrating copy.	Parent cycle, phase, summary, constructive/shadow expressions.	Cannot publish without valid parent.
astrology_modifier	House, domain, uncertainty, or annual-layer modifier.	Eligibility requirements, allowed combinations, precedence.	Validated against combination rules.
astrology_prompt	Reflection or action prompt.	Theme/domain, intensity, exclusions.	Selectable only from approved set.
astrology_safety_rule	Prohibited frames and fallback content.	Rule ID, trigger class, action, reviewed fallback.	Bundled into validation policy.
astrology_release	Editorial release record.	Semantic version, content hashes, approver, changelog, status.	Creates signed bundle and publication webhook.

Release pipeline

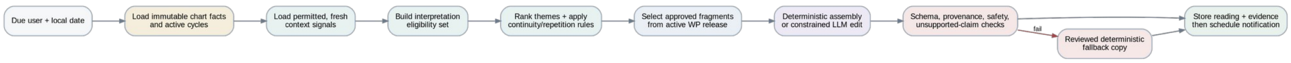
1. Author or edit a content object in WordPress.com.
2. Run field, reference, eligibility, prohibited-claim, and version checks.
3. Require a reviewer distinct from the last author for release approval.
4. Generate canonical JSON, sort keys, compute object and bundle hashes, and sign the release.
5. Send the bundle and signature to the internal Cloudflare release endpoint.
6. Store the immutable bundle in R2, run fixture/safety smoke tests, and atomically activate the release pointer.
7. Retain previous releases for rollback and reading reproducibility.

Hard boundary

WordPress.com may know which editorial objects exist and who approved them. It may not receive a user ID, birth profile, chart, journal, relationship record, health signal, reading, or feedback record.

8. Core workflows

Daily reading pipeline



Reading generation is a governed, idempotent workflow with deterministic fallback.

Workflow inventory

Workflow	Trigger	Major steps	Idempotency key
NormalizeBirthAndCalculateChart	Birth profile create/update	Geocode, historical zone, UTC, Swiss calculation, derivation, snapshot.	user_id + profile_version
RefreshCycleHorizon	Chart publish or daily schedule	Scan horizon, root-refine exact passes, upsert cycles.	chart_fingerprint + horizon_version
GenerateDailyReading	Due-user queue	Load facts/context/release, eligibility, rank, compose, validate, store.	user_id + local_date + release_version
PublishContentRelease	Signed WordPress webhook	Verify, store, fixture test, index, activate/rollback.	release_version + bundle_hash
RefreshConnector	Schedule/provider callback/user action	Validate consent, fetch narrow source, normalize, expire previous.	connector_id + source_window
ExportAccount	User request	Freeze snapshot, collect, decrypt, package, write R2, notify.	export_request_id
DeleteAccount	User confirmation	Freeze jobs, revoke tokens, delete rows/objects, destroy DEK, audit.	deletion_request_id

Daily scheduler

A scheduled Worker queries users whose next_due_at is within the current window and publishes opaque user/date jobs to a Queue. The consumer starts Workflows. Queue delivery is at least once, so the reading uniqueness key is enforced in D1 before any notification is sent.

9. Data architecture

Core records

Entity	Key fields	Sensitive payload	Retention / notes
users	id, status, locale, timezone, created_at	None beyond account metadata.	Until deletion.
user_keys	user_id, key_version, wrapped_dek	Wrapped data-encryption key.	Destroy on deletion.
birth_profiles	user_id, version, accuracy, status	Date/time, place, coordinates, corrections.	Replaced versions retained only for audit policy.
chart_snapshots	id, user_id, fingerprint, contract, container digest	Normalized chart facts and uncertainty.	Immutable; active plus lineage.
cycle_instances	id, chart_id, technique, start/exact/end, phase	Personal timing facts.	Active plus reading lineage.
context_signals	id, source_id, allowed_uses, freshness, consent	Signal value and source detail.	Source-specific; enforced by registry.
content_releases	version, bundle_hash, active_at	Editorial only.	Indefinite for reproducibility.
daily_readings	id, user_id, local_date, release, status, hash	Generated reading text.	User-controlled; lineage immutable while retained.
reading_sources	reading_id, paragraph_id, fact/content/context IDs	Evidence relationships.	Same life as reading.
connector_accounts	provider, scope, token_ref, status, expires_at	Token reference and provider account ID.	Until revoke/delete.
audit_events	actor, action, resource, result, timestamp	No private payload.	13 months minimum target.

Envelope encryption and deletion

```
user record plaintext
  -> AES-GCM with random per-record nonce and per-user DEK
  -> ciphertext stored in D1/R2

per-user DEK
  -> wrapped by versioned root KEK held as a platform secret
  -> wrapped_dek stored in D1

account deletion
  -> revoke connectors
  -> delete active records and objects
  -> destroy wrapped_dek
  -> backup ciphertext becomes cryptographically unrecoverable
```

R2 object layout

```
content/releases/{version}/{bundle_hash}.json
content/renders/{content_id}/{version}.svg
charts/{user_partition}/{chart_id}/{fingerprint}.json.enc
exports/{user_partition}/{request_id}.zip.enc
support/redacted/{case_id}/{artifact_id}
```

10. API and reading assembly

Public API surface

Method	Route	Purpose	Notes
POST	/v1/birth-profiles	Create/update profile and start calculation.	Idempotency required; returns workflow status.
GET	/v1/chart	Normalized chart and uncertainty summary.	No engine-specific raw structures.
GET	/v1/readings/today	Daily reading for user local date.	Stable after publication unless explicitly regenerated.
GET	/v1/readings/{id}/evidence	Paragraph-level provenance.	Facts, content versions, context labels, ranking reasons.
GET	/v1/pattern	Stable natal chapters.	Paginated/versioned.
GET	/v1/timing	Active/upcoming cycles.	Supports phase/domain/duration filters.
GET	/v1/time-travel?date=	Date reconstruction.	Separates historical facts and saved context.
POST	/v1/check-ins	Explicit short-lived context.	User-owned; configurable expiry.
POST	/v1/readings/{id}/feedback	Structured usefulness feedback.	No validity claim inferred from engagement.
GET/PUT	/v1/context-sources	Inspect/modify source permissions.	Allowed uses are explicit.
POST	/v1/exports	Start portable account export.	Short-lived encrypted R2 artifact.
DELETE	/v1/account	Start deletion workflow.	Confirmation and cooling-off policy configurable.
POST	/internal/content-releases	Receive signed WordPress bundle.	Internal service authentication only.

Reading assembly contract

```
{
  "reading_key": "user:<id>:2026-07-30:release-12",
  "facts": [{"id":"cycle_981","phase":"building","orb":0.42}],
  "approved_fragments": ["transit.saturn.square.moon@3/building", "domain.caregiving@2", "prompt.boundaries@4"],
  "context": [{"signal_id":"priority_caregiving","allowed_use":"life_domain_selection"}],
  "output_schema": "daily-reading-v2"
}
```

AI boundary

- Deterministic assembly is the default and must cover every launch configuration.
- A model may improve transitions, compress, or vary phrasing only inside the approved fragment set.
- Private AI Gateway logging is disabled; application telemetry stores route/model, latency, token counts, validation result, and hashes only.
- Any unsupported fact, changed date, diagnosis, fatalism, or source laundering rejects the output and selects reviewed fallback copy.

11. Security, privacy, and governance

Threat / failure	Preventive control	Detective control	Recovery
Birth or journal exposure	Envelope encryption; least privilege; no payload logs.	DLP tests, access audit, canary records.	Revoke access, rotate KEK, incident workflow.
Cross-user access	Central authz policy; opaque IDs; tenant filters in every query.	Authorization integration tests and audit events.	Disable route, invalidate sessions, review affected IDs.
Queue duplicate	Idempotency key and unique constraint.	Duplicate attempt metric.	Return existing result; never send second notification.
WordPress compromise	Signed bundles; read-only runtime release; no user data in WP.	Signature failure alerts and bundle hash audit.	Reject bundle; rollback active release.
Model hallucination	Eligibility allowlist; structured output; deterministic validator.	Validation failure telemetry and sampled red-team tests.	Fallback to reviewed deterministic copy.
Connector overreach	Narrow OAuth scopes; allowed-use policy; token refs only.	Scope drift checks and consent audit.	Pause connector; revoke token; expire signals.
Support misuse	Time-limited audited support roles; redacted artifacts.	Access logs and review queue.	Revoke role, notify, investigate.
Deletion recoverability	Destroy per-user DEK and active data.	Deletion proof record without payload.	Retry workflow; manual escalation.

Service boundaries

Boundary	Allowed	Forbidden
WordPress.com	Editorial objects, release metadata, public content.	All application user data and feedback.
Analytics Engine/logs	Latency, result class, source error, opaque job ID, counts.	Prompt/response, birth data, journal, OAuth body, relationship detail.
LLM provider	Fact IDs, approved content, minimal framing signal, output schema.	Raw device/health data, full calendar, raw journal, stable direct identifiers.
Fly.io standby	Calculation request or encrypted batch input when enabled.	Unencrypted user exports or direct public database access.

Backup deletion rule

D1 Time Travel and other backups can retain historical ciphertext. Per-user crypto-shredding is therefore part of the deletion contract, not an optional optimization.

12. Reliability and observability

Objective	Launch target	Primary measurement	Failure response
API availability	99.9% monthly	Cloudflare requests, 5xx, dependency result class.	Degrade optional context and AI before core readings.
Daily reading readiness	99% before notification window	Workflow completion by user local date.	Retry; deterministic fallback; suppress stale notification.
No duplicate published reading	100%	Unique user/local-date/release key.	Return existing record and acknowledge duplicate job.
Calculation reproducibility	Golden fixtures byte-stable	Container digest, engine/data hashes, chart fingerprint.	Block release on diff until reviewed.
Content rollback	< 10 minutes	Release pointer and drill timestamp.	Activate prior tested bundle.
Source revocation	Before next read/job	Consent check at ingestion and assembly.	Expire signal; cancel pending connector jobs.
Account deletion	< 24 hours	Workflow completion and DEK destruction proof.	Retry and escalate; keep account frozen.

Telemetry contract

- Operational metrics: route, status, latency, queue attempts, workflow step, calculation version, release version, source error class, validation result, and cost buckets.
- Never record prompt text, model response text, birth data, chart prose, journal text, provider OAuth payloads, or relationship details in shared telemetry.
- Use sampled redacted support artifacts only after user action or documented incident procedure.

Disaster recovery

- D1 point-in-time recovery is an operational safeguard, not a substitute for immutable release/artifact storage.
- R2 stores content releases and calculation artifacts by content hash; active pointers can be rebuilt.
- OCI images and Swiss data hashes are retained per release; Fly deployment verifies portability.
- Quarterly drills: content rollback, failed queue replay, database restore, calculation image rollback, and account-deletion retry.

13. Scale and migration triggers

Observed condition	Decision	Why
Cloudflare Container start/compute latency misses the user SLO.	Keep one or more Fly Machines warm or route scheduled bulk work to Fly.	Preserves the same calculation image while improving predictability.
D1 shard size, query shape, or reporting needs exceed the launch model.	Introduce Fly Managed Postgres behind a Fly-hosted data service.	Avoid exposing a private DB and keep Cloudflare clients on an API contract.
Editorial retrieval becomes a bottleneck.	Cache active bundle in KV/R2; use Vectorize only for approved content retrieval.	Never vectorize raw private journals by default.
Provider cost or outages materially affect optional synthesis.	Use AI Gateway dynamic routes, budgets, fallbacks, and version rollback.	Model choice remains operational, not semantic authority.
Connector backlogs or rate limits become persistent.	Split queues by connector, add delayed retries/circuit breakers/DLQs.	Prevents one provider from blocking core reading generation.

Deployment profiles

Profile	Cloudflare	WordPress.com	Fly.io
Launch	All core runtime, data, calculation, orchestration.	Editorial and public site.	CI portability test only.
Warm compute	API/data/orchestration.	Editorial and public site.	Always-warm Swiss service and backfills.
Relational scale	Edge/API/cache/queues; optionally some Workflows.	Editorial and public site.	Data service + Managed Postgres + calculation workers.

14. Delivery sequence and acceptance criteria

Milestone 0 - contracts and licensing

- Resolve Swiss Ephemeris licensing before public activation.
- Freeze calculation, orb, uncertainty, content, evidence, consent, and encryption schemas.
- Create monorepo, environments, CI, signed OCI build, secret policy, and golden-fixture harness.

Milestone 1 - chart integrity

- Birth onboarding and historical time-zone normalization.
- Swiss container, unknown/approximate-time mode, chart fingerprint, natal patterns, and calculation evidence.
- Account, privacy center, export/delete skeleton.

Milestone 2 - editorial control plane

- WordPress.com plugin, custom content types, roles, validation, approval, signed releases.
- Cloudflare ingestion, R2 bundle storage, activation, rollback, and fixture tests.

Milestone 3 - daily product loop

- Cycle horizon, exact passes, ranking, continuity, deterministic assembly, evidence drawer.
- History, notifications, feedback, operational dashboards, failure drills.

Milestone 4 - explicit context and Time Travel

- Priorities, check-ins, life events, journal themes, source controls.
- Free/busy, weather/daylight, date reconstruction, source provenance.

Milestone 5 - optional intelligence and relationships

- Constrained model editing after deterministic coverage is measured.
- Advanced timing layers, semantic editorial retrieval, Bonds after individual quality stabilizes.

Launch acceptance criteria

- Golden chart fixtures match expected normalized facts and fingerprints under the pinned image.
- Unknown and approximate birth-time modes suppress or qualify every unstable claim.
- Every published paragraph resolves to calculated facts and approved content, or is explicitly labeled as user context.
- Stale or revoked signals cannot enter eligibility, ranking, prompting, or notification decisions.
- WordPress release activation and rollback are atomic, signed, fixture-tested, and auditable.
- Queue redelivery cannot create a duplicate reading or duplicate notification.
- Model rejection and provider failure produce reviewed deterministic output.
- Export and deletion complete in staging, including connector revocation and DEK destruction.
- No unregistered source or unapproved use can enter the reading pipeline.

15. Registry implementation map

The complete v0.2 YAML registry is normative and contains 139 entries. The table below summarizes status and phase distribution; the artifact itself contains every source definition and implementation mapping.

Status	Phase distribution	Total
excluded	never: 25	25
experimental	experiment: 7; phase_2: 1	8
optional	launch: 2; phase_1: 13; phase_2: 18; phase_3: 3	36
recommended	launch: 20; phase_1: 8	28
required	launch: 32; phase_1: 1	33
restricted	phase_2: 9	9

Category implementation defaults

Category	Entries	Owner	Runtime	Storage
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B. First-party, user-owned sources	24	first-party-context-service	cloudflare-api-worker + workflow	d1: encrypted structured record
C. Device and operating-system sources	17	device-permission-adapter	mobile-client; cloudflare-api-worker receives minimized aggregate	on-device first; d1: encrypted aggregate only when allowed
D. Linked cloud-service sources	20	linked-service-adapter	cloudflare-connector-worker + workflow	d1: encrypted normalized aggregate; secret store: provider credentials/tokens where applicable
E. Public and ambient sources	12	ambient-data-adapter	cloudflare-connector-worker	kv/cache: short-lived lookup; d1: evidence snapshot only when used
F. Derived and inferred features	22	reading-intelligence-service	cloudflare-workflow + deterministic rules package	d1: derived feature and evidence linkage
G. Deliberately excluded sources and uses	24	policy-gate	none	None

Source lifecycle

```
declared in registry
  -> consent and scope granted
  -> normalized ContextSignal created
  -> freshness, consent, and allowed use checked for every job
  -> expired, paused, or revoked signals excluded immediately
  -> deleted under source retention policy
```

pattern_like_astrology_app_data_source_registry_v0.2.yaml and pattern_like_astrology_app_platform_topology_v0.2.yaml are the machine-readable policy inputs for connector allowlists, consent checks, storage routing, retention jobs, and reading validation.

16. Open decisions and official references

Open product and engineering decisions

Decision	Required resolution
Consumer identity	Choose an OIDC/passkey/magic-link implementation. Do not require a WordPress.com account for ordinary users.
Product name	Choose a distinct identity and visual language; do not imitate The Pattern trade dress or copy.
Solar-return location	Birthplace, residence, or actual location must be explicit and versioned.
LLM launch policy	Decide whether constrained editing ships at launch or after deterministic coverage metrics.
Data region	Decide whether users may select or lock data residency.
D1 migration threshold	Set quantitative shard/query/reporting thresholds before launch.
Push delivery	Select APNs/FCM integration and define notification analytics minimization.
Swiss license	Select the applicable licensing path before public activation or distribution.

Official technical references

Ref	Official source	Why it matters	Link
CF-01	**Cloudflare** Containers overview	Native/full-runtime compute integrated with Workers.	https://developers.cloudflare.com/containers/
CF-02	**Cloudflare** Containers connections to Workers bindings	Container access to D1, R2, KV, Durable Objects, and other bindings.	https://developers.cloudflare.com/containers/platform-details/workers-connections/
CF-03	**Cloudflare** Workflows overview	Durable multi-step execution, retries, waits, and state.	https://developers.cloudflare.com/workflows/
CF-04	**Cloudflare** Queues overview	Guaranteed delivery, batching, retries, delays, and dead-letter handling.	https://developers.cloudflare.com/queues/
CF-05	**Cloudflare** Queues delivery guarantees	At-least-once delivery and idempotency requirements.	https://developers.cloudflare.com/queues/reference/delivery-guarantees/

Ref	Official source	Why it matters	Link
CF-06	**Cloudflare** D1 overview	Serverless SQLite-compatible operational database and read replication.	https://developers.cloudflare.com/d1/
CF-07	**Cloudflare** D1 Time Travel and backups	Point-in-time recovery within the documented retention window.	https://developers.cloudflare.com/d1/reference/time-travel/
CF-08	**Cloudflare** R2 data security	Automatic encryption at rest and TLS in transit.	https://developers.cloudflare.com/r2/reference/data-security/
CF-09	**Cloudflare** AI Gateway logging	Prompt/response logging controls and per-request override.	https://developers.cloudflare.com/ai-gateway/observability/logging/
CF-10	**Cloudflare** AI Gateway dynamic routing	Versioned routing, fallbacks, quotas, budgets, and rollback.	https://developers.cloudflare.com/ai-gateway/features/dynamic-routing/
CF-11	**Cloudflare** Secrets Store overview	Account-level encrypted secrets; currently documented as open beta.	https://developers.cloudflare.com/secrets-store/
CF-12	**Cloudflare** Workers Analytics Engine	High-cardinality operational metrics without storing private payloads.	https://developers.cloudflare.com/analytics/analytics-engine/
CF-13	**Cloudflare** Vectorize overview	Optional semantic retrieval for approved editorial content.	https://developers.cloudflare.com/vectorize/
CF-14	**Cloudflare** Vite plugin for Workers	Web app and Worker development in a production-compatible runtime.	https://developers.cloudflare.com/workers/vite-plugin/
WP-01	**WordPress.com** Use your plugins	Custom plugin deployment is available on paid WordPress.com plans.	https://wordpress.com/support/plugins/use-your-plugins/
WP-02	**WordPress.com** REST API	Content, taxonomy, media, users, and authenticated API access.	https://developer.wordpress.com/docs/api/
WP-03	**WordPress.com** OAuth2 authentication	Scoped WordPress.com and Jetpack API authorization.	https://developer.wordpress.com/docs/api/oauth2/

Ref	Official source	Why it matters	Link
WP-04	**WordPress.org** REST support for custom content types	Custom post type and taxonomy exposure through REST.	https://developer.wordpress.org/rest-api/extending-the-rest-api/adding-rest-api-support-for-custom-content-types/
FLY-01	**Fly.io** Fly Machines	Fast-starting VM/container runtime with lifecycle and region control.	https://fly.io/docs/machines/
FLY-02	**Fly.io** Autostop/autostart Machines	Scale-to-idle behavior for variable workloads.	https://fly.io/docs/launch/autostop-autostart/
FLY-03	**Fly.io** Managed Postgres	Managed HA PostgreSQL, backups, encryption, monitoring, and pooling.	https://fly.io/docs/mpg/
FLY-04	**Fly.io** Managed Postgres client configuration	Pooled/direct connections and PgBouncer constraints.	https://fly.io/docs/mpg/client-configuration/
FLY-05	**Fly.io** Private networking	Private service-to-service networking inside Fly organizations.	https://fly.io/docs/networking/private-networking/
FLY-06	**Fly.io** Secrets and Fly Apps	Encrypted application secrets injected at Machine startup.	https://fly.io/docs/apps/secrets/
SE-01	**Astrodienst** Swiss Ephemeris official repository and releases	Pinned engine and data files; official release metadata.	https://github.com/aloistr/swisseph
SE-02	**Astrodienst** Swiss Ephemeris Programming Interface	Calculation flags, houses, polar fallback, and dual-license requirements.	https://www.astro.com/swisseph-download/doc/swephprg.2.10.htm

Implementation note

This is a product and engineering specification, not legal advice. Swiss Ephemeris licensing, privacy law, app-store review, provider terms, security review, and any research protocol require separate review before public launch.